

EDUCATION

PROGRAM	INSTITUTION	% / CGPA	COMPLETION
Dual degree in Biological Engineering	Indian Institute of Technology Madras	7.65	2025
Minor in Computing	Indian Institute of Technology Madras	6.75	2025
XII (CBSE)	Acharya Bala Siksha Mandir, Puducherry	87.4%	2019
X (CBSE)	Acharya Bala Siksha Mandir, Puducherry	10/10	2017

ACHIEVEMENTS

- \$1500 stipend from Google for successful open-source contribution to **TensorFlow organization** through Google Summer of Code, 2022.
- Chosen as **one of 15 individuals** out of **1,200 candidates** to showcase the final project at the Google Summer of Code 2022 Contributor's Day.
- Ranked top 9% (among 150,000 applicants) in JEE Advanced 2020 and top 0.5% (among 1.1 million applicants) in JEE Mains 2020.

RELEVANT COURSEWORK

- GPU Programming
- Fundamentals of DL(NVIDIA DLI)*
- Probability, Statistics, and Stochastic Process
- Functions of Several Variables
- Series and Matrices
- Neural Networks and DL(Coursera)*
- Fundamentals of Deep Learning
- Special Topics in Reinforcement Learning*
- Nonlinear Optimization

PROFESSIONAL EXPERIENCE

Student Developer - TensorFlow, Google Summer of Code May 2022 - Sept 2022

- Designed a **3D Unity Terrain First Person Shooter Game** and integrated an **ESP32 microcontroller** via BLE to work as the player's console.
- Integrated a microcontroller running a **gesture recognition TinyML model** to recognize and convert gestures and rotations within the game.
- Trained and Optimized a **CNN-based** model aimed at **digit recognition** to fit in less than **40 KB** space and run inference within **1.2 seconds**.
- Deployed the optimized CNN model to work on an **Arduino Nano 33 BLE Sense** with **OV7670 camera module** and **ST7735 TFT LCD module**.

Bioinformatics Intern - Regeneron Pharmaceuticals May 2024 - Aug 2024

- Reverse-engineered benchmarking strategies, evaluated and **red-teamed foundational image segmentation models** on multiple datasets.
- Developed a pipeline for **3D thickness mapping** of organs from MR scans, experimented with processing techniques for output consistency.

ML Engineer Intern - Fusic.co, Japan Sept 2023 - Nov 2023

- Implemented **3D camera pose estimation**, Used **ContactHands** and **LightGlue** models in a feature-matching problem for real-time deployment.
- Proposed and implemented **classical computer vision** algorithms to replace existing ML modules to reduce latency for real-time execution.

RESEARCH EXPERIENCE

Large-scale biological simulations of viruses - Guide: Prof. Razvan Marinescu (UCSC) May 2024 - Present

- Implemented and optimized a **Deep Learning model** trained to emulate action potentials between atoms to predict **virus capsid self-assembly**.
- Experimenting and Reviewing new methods and techniques to optimize the model to achieve State-of-the-art results on standard benchmarks.

Transfer Learning on EEG signals - Guide: Prof. Chandrashekar lakshminarayan Dec 2023 - Present

- Learned differential geometry, Implemented, trained and benchmarked **EEGNet**, **DeepSleepNet**, **EEGInception** models on standard datasets.
- Experimenting and Reviewing new methods and techniques to optimize the model to achieve State-of-the-art results on standard benchmarks.

PROJECTS

SCFasterBERT - CS6024 Course Project Feb 2024 - May 2024

- Developed scBERT2, an optimized model that outperforms SOTA models for scRNA-seq data by enhancing scBERT with novel techniques.
- Implemented architectural improvements including **RMS Norm**, **Flash Attention 2.0**, **SiLU**, and **Sparse MoE** to enhance model performance.

Image Captioning and Machine Translation - CS6910 Course Project Feb 2024 - May 2024

- Engineered an image captioning model using a **CNN with NetVLAD** as the encoder and a RNN-based decoder, evaluated by BLEU scores.
- Built an encoder-decoder based machine translation model using **LSTM networks**, and **IndicBERT** for Indian languages and GloVe for English.

Language Translation - Transformer Paper Implementation Apr 2024

- Engineered and trained an transformer model implementation from "**Attention is All You Need**" paper for **English to Spanish** translation.
- Implemented the core components such as the **multi-head self-attention**, encoder and decoder layers, and positional encodings using PyTorch.

Graph Activation — CS6023 Course Project Mar 2023

- Implemented a CUDA program to parallelly process graph nodes to activate and deactivate nodes based on their edges and nearest neighbors.

Personal Projects - Electronics, ML, Game Development, Python Nov 2020 - May 2022

- Designed a **web scraper** to scrape dynamic text data and processed it to find words from a 20 x 20 puzzle to solve a life skills assignment.
- Developed a modern rendition of the **classic 2D paddle Android** game using procedural 2D map generation and C# scripting in Unity.
- Built a voice-to-voice explainer AI assistant through pipelining **Speech-to-Text**, **ChatGPT API** and **Text-to-Speech** models for accessibility.

CO-CURRICULAR ACTIVITIES

Workshop Trainer, Shastra 2024, IIT Madras Jan 2024

- Conducted a **6-hour** practical hands-on workshop titled "GPT from Scratch" covering deep learning and transformers with **50+** registrations.

SKILLS

• **Programming Languages:** Python, C/C++, Bash — **Tools and Softwares:** TensorFlow, PyTorch, HuggingFace, Pandas, Numpy, Matplotlib

*-MOOCs *-Ongoing